```
clc;
clear;
close all;

% Parameters
Fs = 8000;
fc = 1500;
N = 50;

% Normalized cutoff frequency
wc = fc/(Fs/2);

% Generate FIR filters using different windows
b_rect = fir1(N,wc,rectwin(N+1));
b_hamm = fir1(N,wc,hamming(N+1));
b_hann = fir1(N,wc,hann(N+1));
b_black = fir1(N,wc,blackman(N+1));

% Calculate frequency responses
[H1,f] = freqz(b_rect,1,1024,Fs);
H2 = freqz(b_hamm,1,1024,Fs);
H3 = freqz(b_hann,1,1024,Fs);
H4 = freqz(b_black,1,1024,Fs);

% Display filter coefficients
disp('Rectangular Window Coefficients:');
```

```
Rectangular Window Coefficients:
```

```
disp(b_rect(1:5));
```

```
   -0.0118         0    0.0128    0.0103   -0.0058
```

```
disp('Hamming Window Coefficients:');
```

```
Hamming Window Coefficients:
```

```
disp(b_hamm(1:5));
```

```
   -0.0009         0    0.0012    0.0011   -0.0008
```

```
disp('Hann Window Coefficients:');
```

```
Hann Window Coefficients:
```

```
disp(b_hann(1:5));
```

```
   1.0e-03 *

         0         0    0.2009    0.3592   -0.3588
```

```
disp('Blackman Window Coefficients:');
```

```
Blackman Window Coefficients:
```

```
disp(b_black(1:5));
```

```
   1.0e-03 *

         0         0    0.0743    0.1374   -0.1434
```

```
% Plot frequency responses
figure;

plot(f,20*log10(max(abs(H1),1e-6)),'LineWidth',1.2);
hold on;

plot(f,20*log10(max(abs(H2),1e-6)),'LineWidth',1.2);
plot(f,20*log10(max(abs(H3),1e-6)),'LineWidth',1.2);
plot(f,20*log10(max(abs(H4),1e-6)),'LineWidth',1.2);

grid on;

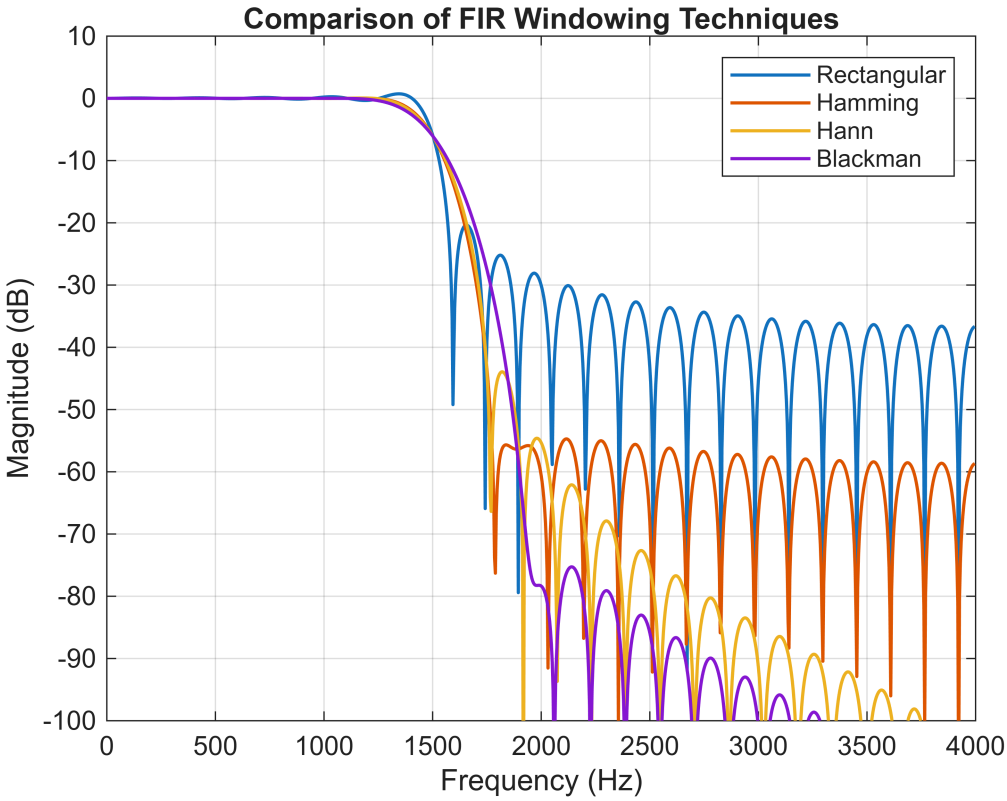
xlabel('Frequency (Hz)');
ylabel('Magnitude (dB)');

title('Comparison of FIR Windowing Techniques');

legend('Rectangular','Hamming','Hann','Blackman');

xlim([0 Fs/2]);
ylim([-100 10]);

hold off;
```




```
disp('FIR Filter Design Completed Successfully!');
```

```
FIR Filter Design Completed Successfully!
```